

CRISPR/dCas12a knock-down of *Acidithiobacillus ferrooxidans* electron transport chain bc₁ complexes enables enhanced metal sulfide bioleaching

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Acidithiobacillus ferrooxidans is an acidophilic chemolithoautotroph that plays an important role in biogeochemical iron and sulfur cycling and is a member of the consortia used in industrial hydrometallurgical processing of copper. Metal sulfide bioleaching is catalyzed by the regeneration of ferric iron; however, bioleaching of chalcopyrite, the dominant unmined form of copper on Earth, is inhibited by surface passivation. Here, we report the implementation of CRISPR interference (CRISPRi) using the catalytically inactive Cas12a (dCas12a) in *A. ferrooxidans* to knock down the expression of genes in the *petI* and *petII* operons. These operons encode bc₁ complex proteins and knockdown of these genes enabled the manipulation (enhancement or repression) of iron oxidation. The *petB2* gene knockdown strain enhanced iron oxidation, leading to enhanced pyrite and chalcopyrite oxidation, which correlated with reduced biofilm formation and decreased surface passivation of the minerals. These findings highlight the utility of CRISPRi/dCas12a technology for engineering *A. ferrooxidans* while unveiling a new strategy to manipulate and improve bioleaching efficiency.

Acidithiobacillus ferrooxidans oxidize iron or reduced inorganic sulfur compounds (RISCs) and have generated interest for use in metal bioleaching as well as additional biotechnology applications (1, 2). In a bioleaching process, metal sulfide ores are oxidized *via* ferric iron and proton attack, either through direct contact mechanisms or indirect interactions, and dissolution can proceed through thiosulfate mechanisms or polysulfide mechanisms (3). The oxidation of many sulfides can be incomplete, as surface passivation can occur. This is especially problematic for the oxidation of chalcopyrite (CuFeS₂), which holds 70% of the unmined copper on Earth, and this material will need to be processed to supply the copper necessary for global electrification. Surface passivation occurs through the formation of jarosite, oxides and polysulfide, and the extent of surface passivation is influenced by biofilm formation (4–6). Given that the RISC metabolism of *A. ferrooxidans* both directly and indirectly influences cell adherence and biofilm formation (7, 8), the

manipulation of cellular metabolism (*i.e.* iron vs RISC utilization) may impact surface passivation, consequently affecting the extent of bioleaching of chalcopyrite and other sulfidic ores.

During aerobic growth, electrons from iron and sulfur are used to reduce O₂ or to produce reducing power (NAD(P)H) through a series of protein complexes (Fig. 1A). The cytochrome bc₁ complex (cytochrome *b* [PetB], cytochrome *c1* [PetC], and the Rieske iron-sulfur protein [PetA]) (9–11) plays an important role in these processes, enabling electron transfer and proton translocation across the membrane, thus contributing to ATP synthesis (10, 12) (Fig. 1B). Under variations in substrate availability or redox states, electron flow can be directed to different pathways, optimizing energy production and metabolic efficiency (13). This electron bifurcation allows the organism to adapt to changing environmental conditions and optimize energy utilization. Due to the various oxidation states of RISCs and the combined involvement of both enzymatic and nonenzymatic reactions in sulfur oxidation (1, 14), the electron fluxes in *A. ferrooxidans* in environments rich in both iron and reduced sulfur, such as sulfidic ores and acid mine drainage (15, 16), are expected to be complex.

In the most studied *A. ferrooxidans* type strain (ATCC 23270), two distinct cytochrome bc₁ complexes are expressed (Fig. 1). The bc₁ complex proteins of the *petI* operon are expressed under iron-rich conditions, while the bc₁ complex proteins of the *petII* operon are expressed under sulfur-rich conditions (11, 17). These two operons are necessary in autotrophic growth to segregate the transfer of electrons to either an “uphill” pathway for the reduction of NAD⁺ or to the “downhill” pathway for the reduction of oxygen, and these two operons are regulated by different promoters and transcriptional regulators, ensuring that each bc₁ complex is expressed under appropriate environmental conditions. During iron oxidation, most of the electrons flow downhill for oxygen reduction, while the *petI* operon enables “uphill” electron transfer to the quinone pool. Under high sulfur concentrations, *A. ferrooxidans* prioritizes the use of sulfur over iron, due to the higher energy density of sulfur (18, 19) (Fig. 1C). Electrons are preferentially transferred “downhill” from the quinone pool by the *petII* bc₁ complex. Thus, iron oxidation is inhibited under sulfur-rich conditions, and this can negatively

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CRISPRi knockdown of *bc₁* complexes in *A. ferrooxidans*

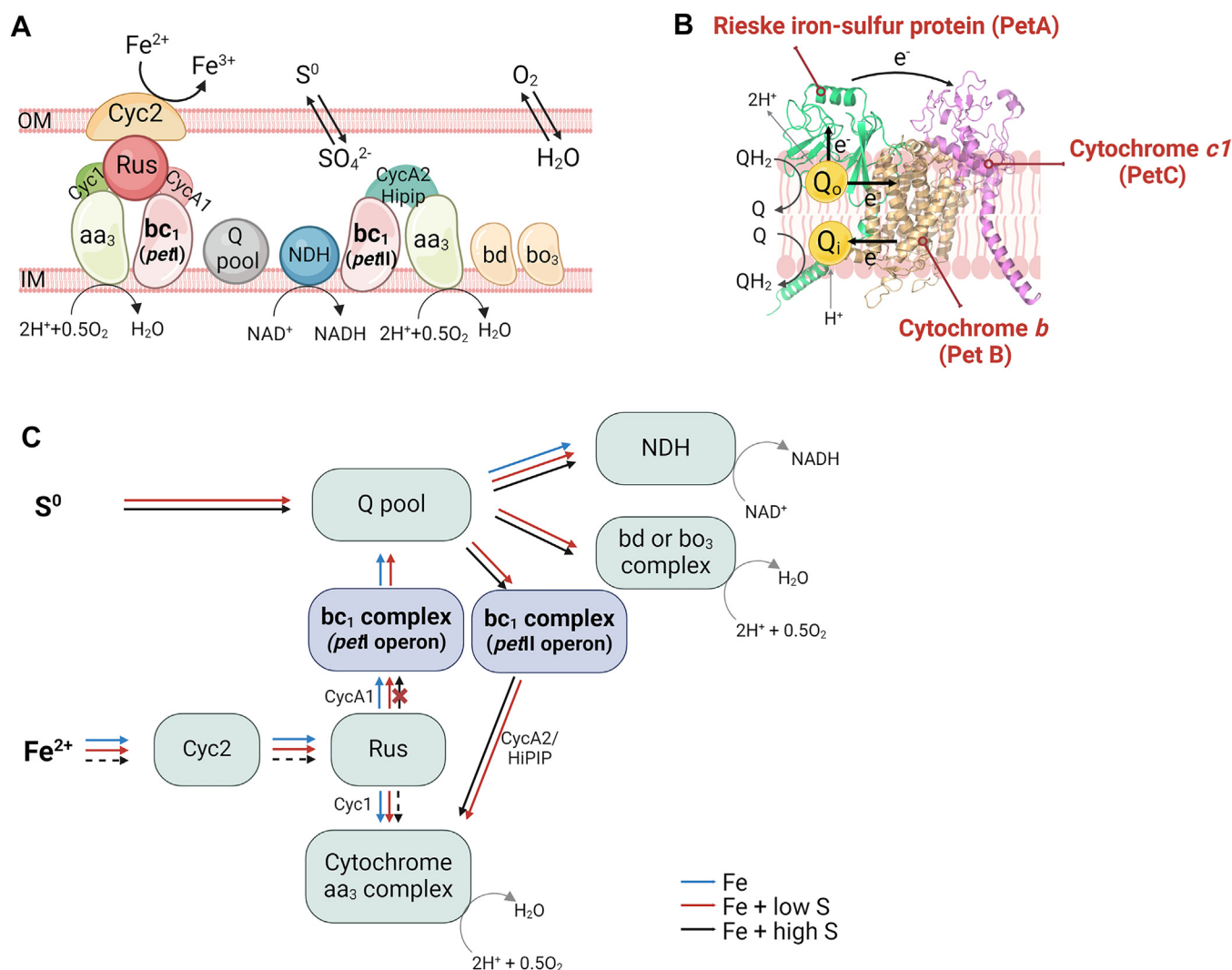


Figure 1. Overview of iron and sulfur oxidation proteins and electron transport pathways of *A. ferrooxidans*. A, protein complexes implicated in iron and sulfur oxidation, (B) structural diagram of the *bc₁* complex, and (C) electron fluxes that occur during iron and sulfur oxidation under different conditions. In Panel B, an electron is serially transferred to Rieske iron-sulfur protein (PetA, Uniprot identification Q93A06) and cytochrome *c*1 (PetC, Uniprot identification Q93A10), as ubiquinol is oxidized at Q_o site. Simultaneously, another electron travels back across the membrane through cytochrome *b* (PetB, Uniprot identification Q9KIW3), contributing to the reduction of quinone at the Q_i site. In Panel C, the electrons from the oxidation of iron and RISCs are transmitted either to reduce O_2 or NAD^+ via electron carrier proteins, in the presence of iron (blue), iron and low sulfur (0.1%, w/v) (red), and iron and high sulfur (0.5%, w/v) (black, hypothetical). During iron oxidation, approximately 95% of electrons are transported to downhill pathway (Cyc2 → Rusticyanin (Rus) → Cyc1 → aa₃ oxidase) and 5% of electrons are passed uphill pathway (Cyc2 → Rus → CycA1 → *bc₁* complex (encoded by *petI* operon) → Quinone pool (Q pool) → NADH hydrogenase (NDH)). During sulfur oxidation, the electrons are directly transmitted to enzymes (Q pool → *bc₁* complex (encoded by *petII* operon) → (Hipip and CycA2) → aa₃ oxidase), or bd/bo₃ complexes), or directly to NDH for NAD^+ reduction. Arrows with dashed or X marks indicate lower or inhibited electron flows, respectively. IM, inner-membrane; Q_i, quinol-reduction site; OM, outer-membrane; Q_o, quinone oxidation site. The Figure was created by Biorender.com.

affect the bioleaching of iron sulfide minerals including chalcopyrite. We hypothesized that the down-regulation of the *petII* operon may enhance iron oxidation under high sulfur concentrations, offering potential opportunities for influencing bioleaching mechanisms, leading to reduced surface passivation and enhancing metal sulfide bioleaching.

The use of Clustered Regularly Interspaced Short Palindromic Repeats (CRISPR) and CRISPR-associated protein (Cas) system for genome editing has revolutionized genetic engineering (20). CRISPR interference (CRISPRi) uses a catalytically inactive (nuclease-deficient) Cas protein to silence target genes, enabling functional studies of genes without

genome modification. Although extensively applied in many bacteria, CRISPRi applications to acidophiles, including *A. ferrooxidans*, have been limited. A Type IV CRISPR/Cas was identified in *A. ferrooxidans* ATCC23270 (21), but this endogenous system remains uncharacterized. Recently a well-established Class 2 Type II CRISPR/nuclease-deficient Cas9 (dCas9) was introduced to *A. ferrooxidans* (DSM 14882), and the authors used this CRISPRi to knock down the expression of the nitrogenase *nifH* gene, observing decreased growth on ammonium (NH_4^+) (22). Another study used dCas9 in *Acidithiobacillus ferridurans* to decrease sulfur oxidation by suppressing the expression of HdrA (heterodisulfide reductase)

and TusA (thiosulfate carrier) (23). These reports demonstrate the successful transcriptional repression of target genes; however, CRISPRi has yet to be used to create cells with useful new attributes. Moreover, the high off-target efficiency and cytotoxicity of Cas9 can restrict its use in many bacteria (24, 25). To address this, another Class 2 system, Cas12a (Cpf1) has become a useful alternative with functional advantages, such as easy multiplex genome editing, compatibility with high GC organisms (such as *A. ferrooxidans*), and lower toxicity (25–28). However, the use of Cas12a based CRISPRi systems in non-model bacteria, including extremophiles, has been rarely reported.

Here, we report CRISPRi with DNase-deactivated *Francisella novicida* Cas12a (dCas12a) in *A. ferrooxidans*. After optimizing the CRISPRi system in *Escherichia coli* for enhanced knockdown efficiency, this system was investigated in *A. ferrooxidans* to reduce expression of the *petA2* and *petB2* genes (in the *petII* operon), and the *petA1* and *petB1* genes (in the *petI* operon). The mutant cells responded differently depending on available sulfur concentrations and the cells were further evaluated for their impact on the bioleaching of the sulfidic ores, pyrite and chalcopyrite, as well as the impact of these mutations on biofilm formation and surface passivation.

Results

dCas12a constructed in pJRD vector was evaluated in *E. coli*

The catalytically active FnCas12a (29) was cloned into pJRD215, a broad-host-range mobilizable vector that has been

used for the transformation of *A. ferrooxidans* (1) (Fig. 2A). The expression of the Cas gene was controlled by *tac* promoter, which is a strong constitutive promoter effective for gene expression in *A. ferrooxidans* (30). The FnCas12a was converted to a catalytically inactivated form (dCas12a) by introducing the D917A/E1006A double mutations, inactivating the DNase activity without interrupting the RNA processing and DNA binding (29, 31).

We inserted the dCas12a following the J23119 promoter up to the first direct repeat from pFnCpf1_min (29), into pJRD215 vector, in which the expression of the designed dCas12a is controlled by *tac* promoter (Fig. 2A). The *lacY* gene was chosen as a target for knockdown in *E. coli* cells as it can be used as a growth-dependent phenotype. As the design of CRISPR RNA (crRNA) affects RNA processing and targeting efficiency (32), the arrays were designed with one target spacer (20 bp) with different sequences in placY1 and placY2, and with two target spacers in placY3 (Fig. 2B). For placY1 and placY2 plasmids, additional sequences (TTTTTTGTCTAGCTTTAATGC) from pFnCpf1_min were included to ensure sufficient physical space for dCas12a to process the designed crRNA, avoiding potential interference by *rrnB*. In addition, this additional sequence was slightly modified in placY3 (TTTCCAGAACTATTTAATGC) to maximize free energy of secondary structure. Structural analyses were performed by NUPACK, and the placY1 showed secondary structures (Fig. 2C), which can potentially disrupt the proper operation of the CRISPR array. The placY2 showed better RNA structure with proper folding of direct repeats, while the placY3 further improved the structures, resulting in a

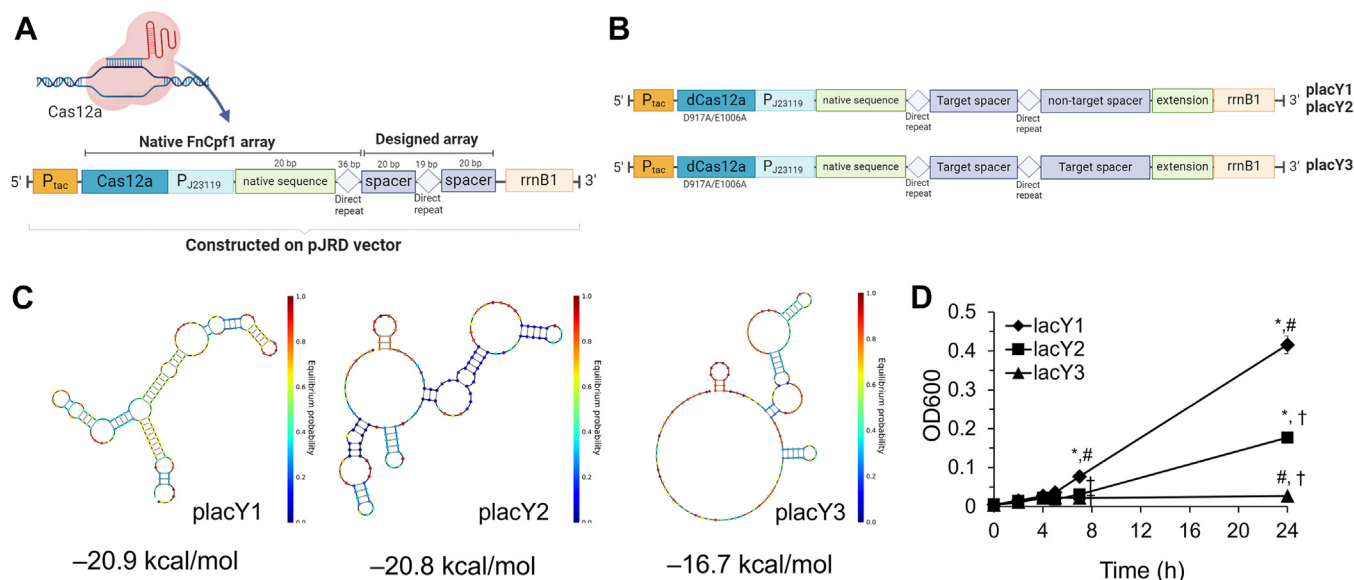


Figure 2. gRNA design for dCas12a CRISPRi in *A. ferrooxidans*. A, a schematic diagram of the dCas12a system constructed in the pJRD215 vector. The catalytically active FnCas12a (29) and the J23119 promoter up to the first direct repeat sequences were amplified from pFnCpf1_min, followed by inactivation of Cas12a and insertion of spacers, then cloned onto pJRD215 vector with *rrnB* terminator. The FnCas12 was made catalytically inactive by D917A/E1006A mutations (dCas12a), and the expression of the designed dCas12a system is controlled by *tac* promoter. B, different designs of dCas12a arrays with one target spacer with different sequences (placY1 and placY2), and two target spacers (placY3). C, predicted secondary structures of the different dCas12a designs with placY1, placY2, and placY3 formulated by NUPACK webserver (<http://www.nupack.org/>) (59, 60). The free energy of the secondary structure was estimated. D, growth of *E. coli* cells (lacY1, lacY2, and lacY3) transformed with placY1, placY2, and placY3 plasmids, respectively, under lactose, in terms of OD600. Error bars indicate the standard deviations of triplicate biological replicates. Symbols indicate statistical significance ($p < 0.05$): *, compared to lacY1; #, compared to lacY2; †, compared to lacY3.

higher free energy. Then, these different dCas12a arrays were evaluated for growth suppression of *E. coli* cells. Under lactose conditions, the growth of *E. coli* transformed with lacY2 was inhibited by almost half as compared to the cells with lacY1, while the cells with lacY3 ceased growing before reaching the exponential phase (Fig. 2D). These results indicated that the use of two target spacer sequences, while minimizing the secondary RNA structure, increased the on-target efficiency. We opted to use the two target spacers without additional sequence extension (before *rrnB* terminator) for the subsequent experiments, since the proper functioning of the designed CRISPR array in the absence of additional sequences was observed, in terms of knockdown efficiency, compared to the number of target spacers (data not shown). These results confirm that the dCas12a expressed *via* the pJRD215 vector reduced the transcription of the target gene without apparent toxicity to *E. coli* cells, which is important considering that plasmid transfer from *E. coli* to *A. ferrooxidans* is required for conjugal transformation.

Knockdown of *petA2* and *petB2* overcome suppressed iron oxidation in *A. ferrooxidans*

After confirming the functionality of the dCas12a expressed from the pJRD215 plasmid in *E. coli*, the plasmids were conjugally transferred to *A. ferrooxidans*. The *petA2* and *petB2* genes in *petII* operon, and *petA1* and *petB1* genes in *petI* operon, which are two highly expressed genes under growth in sulfur and iron, respectively (17), were targeted for knockdown (Fig. 1). The engineered strains were referred to as dPetA2, dPetB2, dPetA1, and dPetB1. In the absence of sulfur (Fe-only medium), the *petA1* and *petB1* genes were expressed at a higher level than the *petA2* and *petB2* genes, confirming that these genes are related to iron oxidation. The transcriptional expression levels of the targeted genes in each of the engineered cells were decreased by ~90% as compared to the wild-type cells (Table 1). Overall, these knockdowns did not affect

the expression levels of the non-targeted genes, although the expression of *petA2* was increased compared to the WT in the dPetB1 strain.

When the cells were grown in iron and low sulfur media (0.1%, w/v), the expression levels of all four of the measured genes were similar in the wild type cells. And again, transcriptional expression levels of the targeted genes in each of the engineered cells were decreased by ~90% as compared to the wild type cells. This contrasted with the iron and high sulfur conditions (0.5%, w/v), where the *petA2* and *petB2* genes were expressed at a higher level than the *petA1* and *petB1* genes. In these conditions, the transcriptional expression levels of the *petA2* and *petB2* genes in the engineered cells were decreased by ~90% as compared to the wild type. However, in the dPetA1 and dPetB1 knockouts, the lower levels of expression of the iron related *petA1* and *petB1* were not further suppressed, and in the dPetA2 and dPetB2 knockouts, increased levels of expression of *petA1* and *petB1* were observed.

Overall, these results demonstrate the successful knockdown of the targeted genes in *A. ferrooxidans* with low off-target efficiency by the dCas12a system constructed in the pJRD plasmid. The *tac* promoter successfully drove the expression of the dCas12 and when combined with the J23119 promoter, the design was sufficient to support the expression of downstream designed CRISPRi array gRNA sequences in *A. ferrooxidans*, as has been observed in other extremophiles (33–35).

The engineered cells were grown under different iron and sulfur concentrations, and their phenotypic responses were compared to the wild-type cells. Under the high sulfur conditions (0.5%, w/v), the wild-type cells oxidized sulfur while the iron oxidation was found to be repressed (Fig. 3, A and B), which is consistent with the low expression of the *petA1* and *petB1* genes. This was in contrast to the dPetA2 and dPetB2 strains which oxidized Fe²⁺, and less sulfur was oxidized by both cell lines (Fig. 3, A and B). The effect was more pronounced in the dPetB2 cells. These results suggest that the

Table 1

Transcriptional gene expression of *petA2*, *petB2*, *petA1*, and *petB1* genes in the wild type (WT) and engineered *A. ferrooxidans* with knock-down *petA2* (dPetA2), *petB2* (dPetB2), *petA1* (dPetA1), and *petB1* (dPetB1) genes under different growth conditions

Cells	Transcriptional gene expression (× 10 ⁸ copies/g cDNA)			
	<i>petA2</i>	<i>petB2</i>	<i>petA1</i>	<i>petB1</i>
Fe only				
WT	3.2 ± 0.3	2.1 ± 0.5	9.9 ± 0.2	8.6 ± 1.0
dPetA2	0.51 ± 0.02*	3.0 ± 0.1	8.7 ± 0.5	9.3 ± 0.9
dPetB2	2.7 ± 0.6	0.13 ± 0.02*	9.3 ± 0.8	9.0 ± 0.4
dPetA1	3.4 ± 0.8	1.9 ± 0.1	0.25 ± 0.04*	8.3 ± 0.1
dPetB1	4.7 ± 0.3*	3.1 ± 0.1	9.1 ± 0.5	0.89 ± 0.06*
Fe and low S (0.1%, w/v)				
WT	8.3 ± 0.3	9.2 ± 0.8	7.6 ± 1.1	7.8 ± 0.5
dPetA2	0.81 ± 0.02*	8.7 ± 1.5	6.7 ± 0.2	7.8 ± 0.2
dPetB2	7.3 ± 1.2	1.0 ± 0.1*	9.1 ± 0.5	8.6 ± 1.1
dPetA1	7.0 ± 0.2*	8.9 ± 1.5	0.73 ± 0.03*	7.7 ± 0.3
dPetB1	8.4 ± 0.7	8.6 ± 0.9	7.4 ± 1.1	0.74 ± 0.12*
Fe and high S (0.5%, w/v)				
WT	9.2 ± 1.1	9.8 ± 1.0	0.84 ± 0.02	0.14 ± 0.03
dPetA2	1.2 ± 0.2*	9.3 ± 1.1	6.8 ± 0.5*	7.2 ± 0.1*
dPetB2	9.5 ± 0.1	0.91 ± 0.05*	6.5 ± 0.2*	8.1 ± 0.01*
dPetA1	8.8 ± 1.0	9.0 ± 0.9	1.0 ± 0.1	0.85 ± 0.02
dPetB1	9.4 ± 0.8	9.6 ± 0.6	0.38 ± 0.01	0.29 ± 0.01

All tested growth medium contains 100 mM Fe²⁺. Symbols (*) indicate statistical significance (*p* < 0.05) of the gene expressions in knockdown cells compared to the wild type cells.

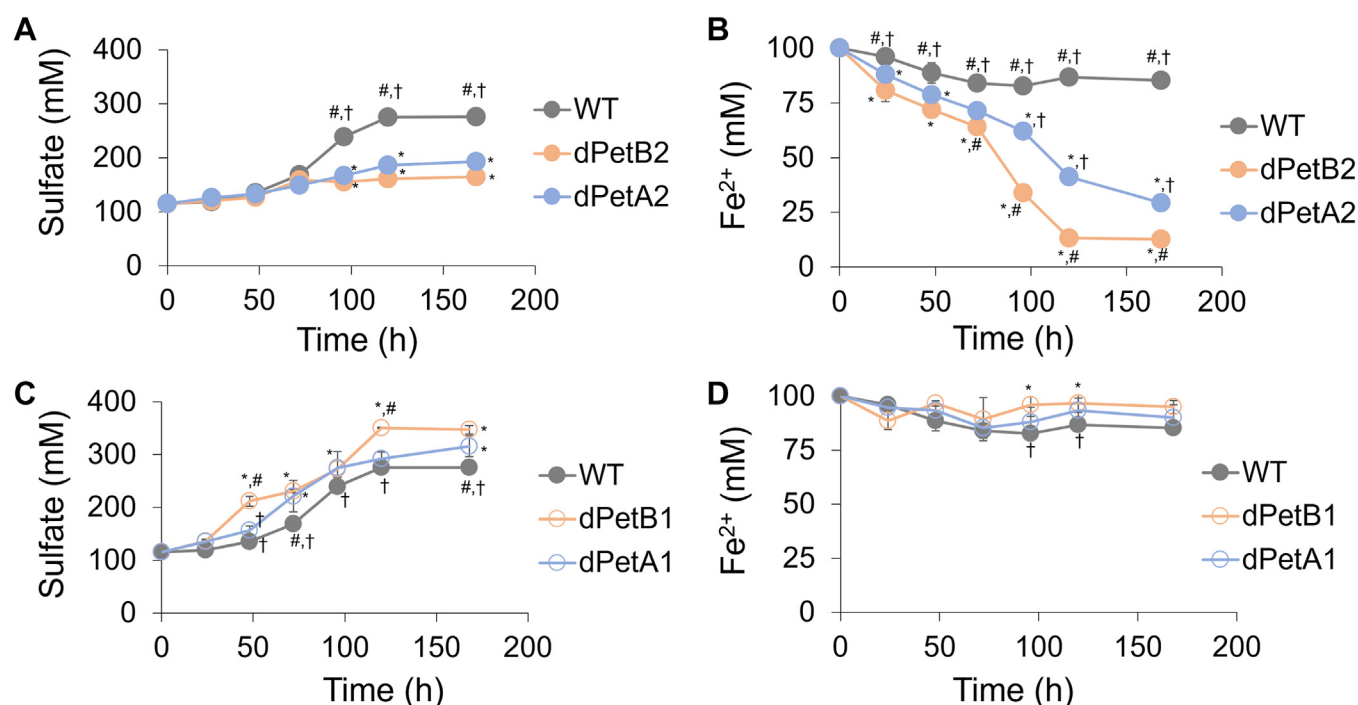


Figure 3. Sulfur and iron oxidation under high sulfur conditions. Profiles of sulfate production and Fe^{2+} consumption of the wild type (WT) and engineered *A. ferrooxidans* with knockdown *petA2* (dPetA2) and *petB2* (dPetB2) genes (A and B), and *petA1* (dPetA1) and *petB1* (dPetB1) genes (C and D), under high sulfur conditions (0.5% S, w/v) over time. Error bars indicate the standard deviations of triplicate biological replicates. Symbols indicate statistical significance ($p < 0.05$): *, compared to WT; #, compared to dPetA2 (for A and B) or dPetA1 (for C and D); †, compared to dPetB2 (for A and B) or dPetB1 (for C and D).

knockdown of either the *petA2* or *petB2* genes, which resulted in an increase in the *petA1* and *petB1* genes, enables the cells to overcome the suppression of iron oxidation that normally occurs under high sulfur conditions (Fig. 1 and Table 1). In contrast, the profiles of iron oxidation were comparable among both dPetA2 and dPetB2 engineered and wild type cells when growing only with iron, which supports the functions of the *petA2* and *petB2* genes for the support of sulfur oxidation (Fig. S1 and Table 1). The knockdowns of the *petA1* and *petB1* genes showed the opposite results, as the iron oxidation remained largely suppressed, while the sulfur oxidation was similar or enhanced as compared to the wild type cells (Fig. 3, C and D). The repressed iron oxidation of both dPetA1 and dPetB1 cells in iron only medium further supports the functional roles of these genes in iron oxidation (Fig. S1 and Table 1).

Different effects were seen under lower sulfur conditions (0.1% S, w/v) (Fig. 4). Sulfur oxidation by the dPetA2 and dPetB2 cells were similar to the wild type while the iron oxidation was even further accelerated in the knockdowns, with dPetB2 again showing the greatest effect (Fig. 4, A and B). However, under these same conditions, the dPetA1 and dPetB1 cells exhibited enhanced sulfur oxidation. The knockdowns of the *petA1* and *petB1* genes led to reduced iron oxidation as compared to the wild type cells, which is not surprising considering the majority of electrons (~95%) from iron transfers *via* the downhill pathway (Figs. 1 and 4, C and D). Overall, these results indicate that *A. ferrooxidans* derives energy preferably from sulfur over iron when both substrates

are present, and this can be manipulated in either direction with the knockdowns of the two different *bc₁* complexes (Figs. 3 and 4).

Enhanced iron oxidation improves bioleaching

A. ferrooxidans has been widely investigated for its role in industrial mineral sulfide bioleaching and it has been shown to participate in direct contact interactions *via* biofilm formation as well as indirect, non-contact interactions by planktonic cells. Given the improved iron oxidation observed under sulfur abundant conditions by the dPetA2 and dPetB2 cells, we examined the effect of these knockdowns on the bioleaching of iron-bearing sulfidic ores, pyrite (FeS_2) and chalcopyrite. Since we observed the better performance of the dPetB2 cells over the dPetA2 cells, in terms of the iron oxidation (Figs. 3 and 4), only the dPetB2 and dPetB1 cells, targeting of the knockdowns of the cytochrome *b*, were evaluated in the bioleaching experiments, while wild type cells served as controls.

Remarkably, dPetB2 cells exhibited enhanced bioleaching efficiencies of both minerals. The dPetB2 cells achieved final iron and copper bioleaching efficiencies of $35 \pm 1\%$ and $68 \pm 3\%$ from pyrite and chalcopyrite, respectively, which are up to 4- and 3-fold higher than the dPetB1 or wild-type control strains (Fig. 5, A and B). The solid residues after the bioleaching experiments were found to be mainly composed of sulfur, jarosite, and unreacted pyrite and/or chalcopyrite, in all conditions (Fig. S2). It is noteworthy that the conditions with the dPetB2 cells had fewer peaks corresponding to jarosite,

CRISPRi knockdown of *bc₁* complexes in *A. ferrooxidans*

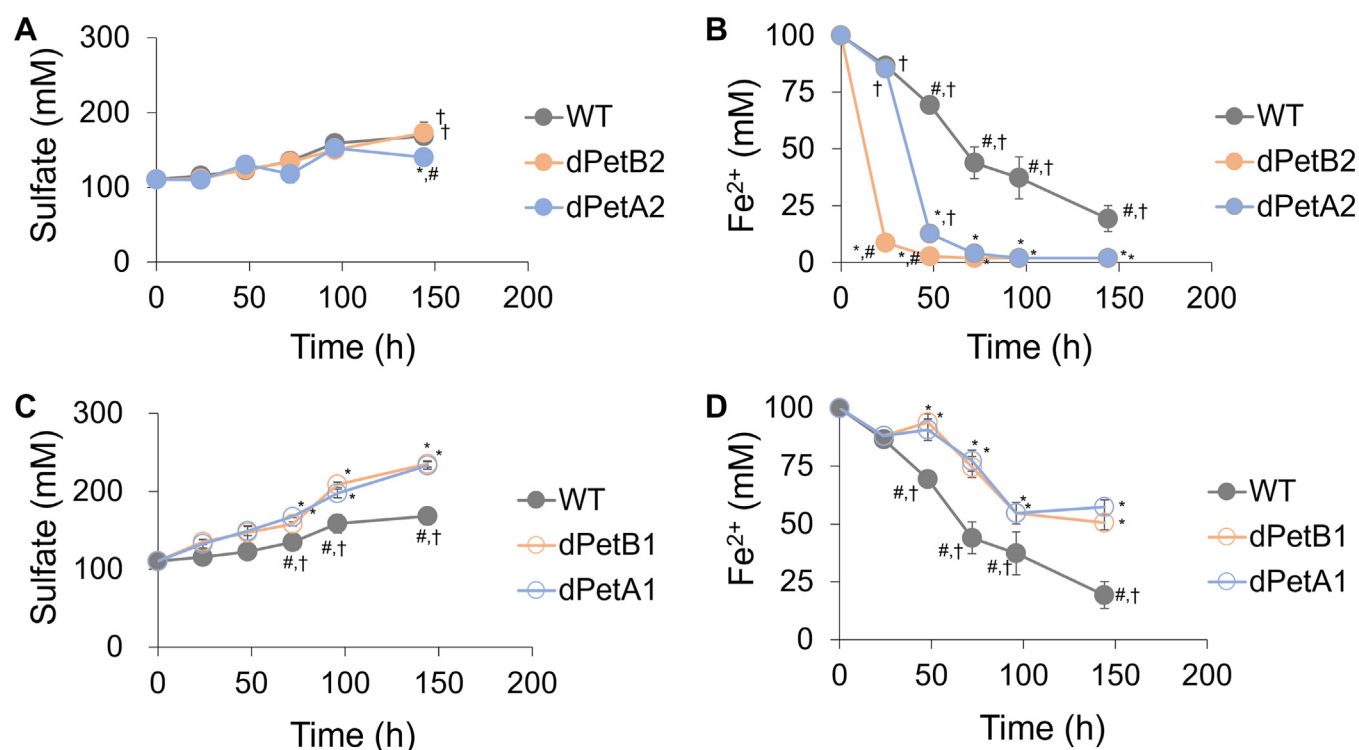


Figure 4. Sulfur and iron oxidation under low sulfur conditions. Profiles of sulfate production and Fe^{2+} consumption of the wild type (WT) and engineered *A. ferrooxidans* with knockdown *petA2* (dPetA2) and *petB2* (dPetB2) genes (A and B), and *petA1* (dPetA1) and *petB1* (dPetB1) genes (C and D), under low sulfur conditions (0.1% S, w/v) over time. Error bars indicate the standard deviations of triplicate biological replicates. Symbols indicate statistical significance ($p < 0.05$): *, compared to WT; #, compared to dPetA2 (for A and B) or dPetA1 (for C and D); †, compared to dPetB2 (for A and B) or dPetB1 (for C and D).

particularly for chalcopyrite, in comparison with the wild type and dPetB1 cells. Considering that the jarosite formation is the main cause of the surface passivation (36, 37), reduced jarosite formation is consistent with the higher bioleaching efficiency observed with this cell line (Fig. 5, A and B). Furthermore, elemental analysis results indicated that the peak corresponding to carbon was not identified in either the pyrite and chalcopyrite residues after the bioleaching with dPetB2 cells, compared to the wild type and dPetB1 cells (Fig. S3). This also implies that fewer dPetB2 cells and biofilm materials were attached to the minerals during the experiments.

The mineral contact modes of the cells were further evaluated, and 2-fold more planktonic cells along with up to 4- and 6-fold less biofilm formation was observed in the pyrite and chalcopyrite bioleaching with the dPetB2 cells, respectively, as compared to the dPetB1 and wild type controls (Fig. 5, C and D). These results were consistent with the significantly lower transcriptional expression of the genes involved in the biofilm formation in *A. ferrooxidans* (i.e., expressions of genes responsible for quorum sensing, cyclic-di-GMP (c-di-GMP), and extracellular polymeric substance) which was observed under the bioleaching conditions for both minerals with the dPetB2 cells (Fig. 5, E–G). In addition, the expression of biofilm genes were higher in the conditions with the dPetB1 cells than the wild type cells, although both conditions showed comparable quantitative biofilm results. Overall, the enhanced bioleaching performance of the dPetB2 cells correlated with a substantial reduction in key matrices of cell attachment, and

the knockout strain demonstrated a remarkable improvement in chalcopyrite bioleaching efficiency, compared to the wild type cells (Table S1).

The knockdown of the cytochrome *b* in the dPetB2 cells caused the cells to reduce biofilm formation and remain in a planktonic state. This appears to play a crucial role in enhancing bioleaching, considering that biofilm formation can promote surface passivation *via* jarosite formation (4, 5). Given that the extracellular sulfur metabolites and proteins generated in sulfur metabolism of *A. ferrooxidans* contribute to cell attachment and biofilm formation (7, 8), we further examined the effect of the reduced sulfur oxidation on bioleaching. This was evaluated in wild type *A. ferrooxidans* cells by removing or adding sulfur intermediates that are produced during sulfur oxidation to chalcopyrite bioleaching experiments. When the sulfur intermediates were removed by periodic replacement of the leaching medium, the copper bioleaching efficiency was increased by 1.4-fold (Fig. 5H). On the other hand, the bioleaching efficiencies deteriorated upon the addition of thio-sulfate and tetrathionate, which are major RISCs generated during the sulfur oxidation, as compared to the controls (Fig. 5I).

Discussion

A. ferrooxidans is rare in its ability to oxidize both iron and sulfur, and the manipulation of this substrate preference will be important as these industrially important microbes are

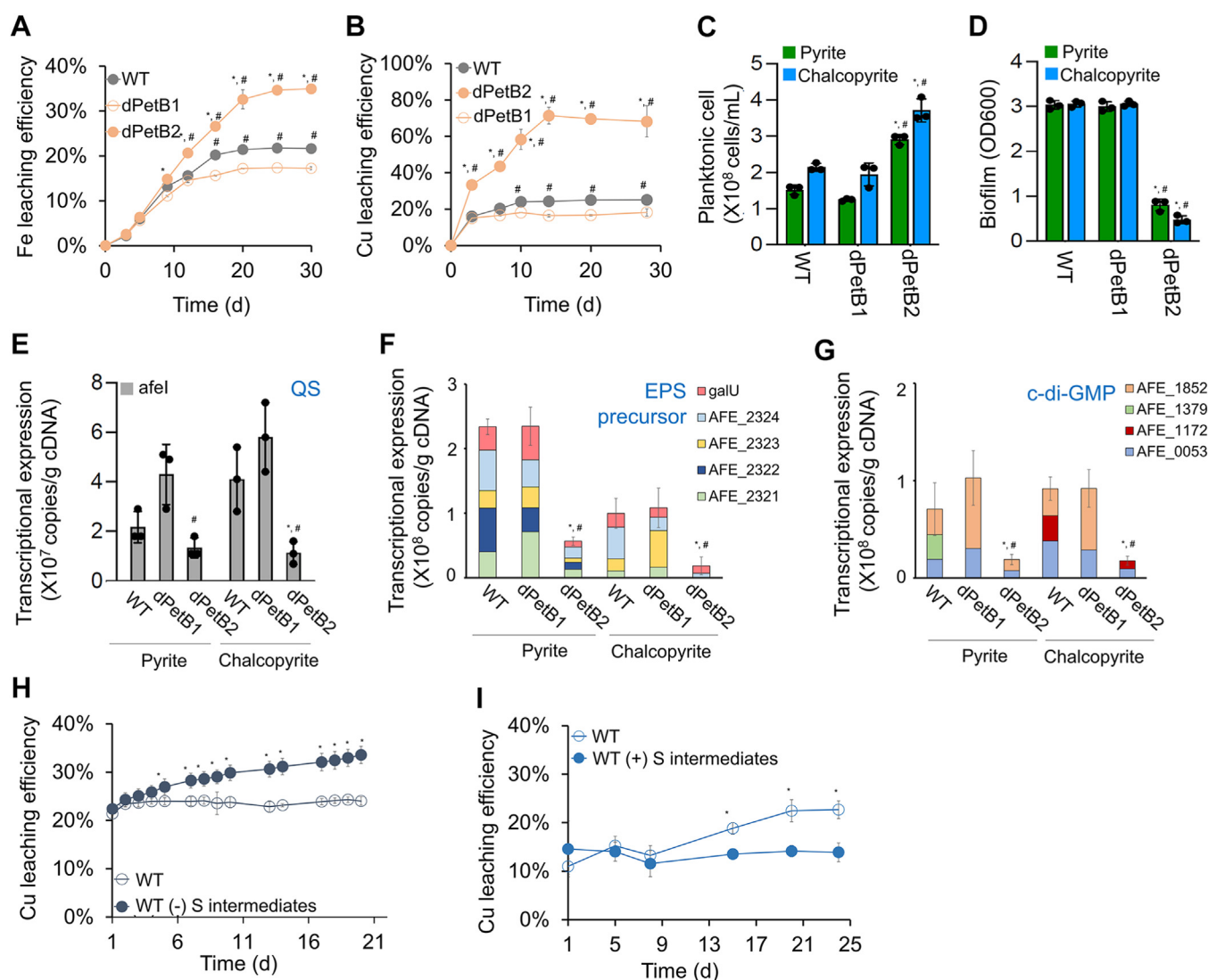


Figure 5. Effects of *petB2* knockdown on bioleaching efficiency. A, iron leaching efficiency from pyrite (FeS₂) of the wild type (WT) and engineered *A. ferrooxidans* with knockdown *petB2* (dPetB2) and *petB1* (dPetB1) genes. B, copper leaching efficiency from chalcopyrite (CuFeS₂) of both wild type and engineered cells. C, quantification of planktonic cells of both wild-type and engineered cells present in the solutions after bioleaching of pyrite and chalcopyrite. D, quantification of biofilm formation of both wild-type and engineered cells from the mineral residues after bioleaching of pyrite and chalcopyrite. E, transcriptional expressions of the genes responsible for biofilm formation by *A. ferrooxidans*, including quorum sensing (QS), (F) extracellular polymeric substance (EPS) precursors, and (G) cyclic-di-GMP (c-di-GMP). H, copper leaching efficiency from chalcopyrite with the WT cells without (WT) or with the removal of sulfur intermediates via periodic replacement of leaching media (WT – S intermediates), and (I) the addition of sulfur intermediates (thiosulfate and tetrathionate, WT + S intermediates) (* indicates the statistical significance ($p < 0.05$) compared to the control). Error bars indicate the standard deviations of at least triplicate analyses from triplicate biological replicates. Symbols indicate the statistical significance ($p < 0.05$) compared to the wild type (*) and dPetB1 cells (#), unless otherwise noted.

developed for biotechnology applications. Although it is known that the *petI* and *petII* operons are found to encode ubiquinol cytochrome *c* reductase *bc₁* complexes, which catalyze electron transport during iron and sulfur oxidation (11, 14, 17) (Fig. 1), a full mechanistic understanding of the roles these operons play in metabolism and substrate selectivity is lacking, partially due to limited genetic tools for these organisms. The transcriptional expressions of the genes encoding PetAB1 and PetAB2 under various growth conditions (Table 1) confirm that the expression of these proteins is influenced by the availability and abundance of energy sources. Using CRISPRi with dCas12a, we observed that the down-regulation of genes *petA2* and *petB2* (in the *petII* operon) led

to improved iron oxidation by *A. ferrooxidans* in the presence of sulfur, particularly under high sulfur conditions where the iron oxidation is most suppressed in the wild type cells (Table 1, Figs. 3, and 4). In contrast, the suppression of *petA1* and *petB1* genes (in the *petI* operon) led to decreased iron oxidation by the cells as compared to the wild type. These results indicate that the energy substrate utilization can be tuned via regulation of the expression of genes in the *petI* and *petII* operons. Moreover, it seems that knockdown of the cytochrome *b* (PetB) had more of an impact than the knockdown of the Rieske iron-sulfur protein (PetA) in both operons, suggesting the cytochrome *b* is a metabolic control point for regulating substrate utilization, given its more direct role in

CRISPRi knockdown of *bc₁* complexes in *A. ferrooxidans*

electron transfer with the quinone pool. In the case of the *petII* operon, the observation that the *petB2* gene plays more a prominent role than *petA2* during sulfur oxidation is consistent with its higher transcriptional gene expression compared to other genes within *petII* operon (17).

It is worth noting how manipulation of the major electron transport gene influences the metabolic pathways within *A. ferrooxidans*. A signal transducing system RegBA has been suggested to bind to the operons involved in iron or sulfur oxidation in *A. ferrooxidans* (38), in response to redox changes in the quinone pool. Variations in electron fluxes could be sensed by the signal transducing system, allowing the cells to establish a hierarchical order of substrate utilization (39, 40). Given that cytochrome *bc₁* complexes bifurcate electron transfer upon substrate availability for energy conservation (13, 41), targeting major genes controlling electron transfer, as opposed to targeting a specific gene responsible for iron/sulfur oxidation, leads to the significant attenuation of overall substrate utilization by single gene knockdowns. The observed metabolic flexibility of *A. ferrooxidans* could also impact the growth and adaptation of the cells to varying environmental conditions, which is consistent with its competitiveness in different ecological niches.

The enhanced iron oxidation of *A. ferrooxidans* under sulfur-abundant conditions can have a significant impact on metal recovery from sulfidic ores, such as chalcopyrite which is the most abundant yet recalcitrant copper sulfide (4). Remarkably, bioleaching with the dPetB2 cells resulted in approximately 70% of the copper being extracted from chalcopyrite, which is substantially better than what has ever been reported for the bioleaching efficiency of this mineral (Table S1) (6, 42, 43). During the bioleaching of chalcopyrite, direct contact by cells in biofilms as well as non-contact by planktonic cells contributes to copper liberation (6). The knockdown of the *petB2* gene led to increased iron oxidation and this correlated with the cell localization during the bioleaching, as these cells exhibited a tendency towards a planktonic state, with lower biofilm formation and related biofilm

gene expression (Figs. 5, C–G and S3). These results suggest that the dominant bioleaching mechanisms can be engineered by regulating the metabolic gene expressions in *A. ferrooxidans*. It is plausible that the promoted iron oxidation combined with the reduced sulfur oxidation spontaneously encourages the activity of the planktonic cells, thus the overall bioleaching was shifted to favor the indirect, non-contact pathway (Fig. 6). The higher cell attachment to minerals under the bioleaching conditions with the dPetB1 cells supports this possibility (Fig. S3). Although the cell attachment is an important step for the mineral bioleaching, the excessive biofilm formation likely promotes surface passivation of chalcopyrite via jarosite formation, which inhibits the bioleaching (6). We observed less jarosite formation under the dPetB2 conditions (Fig. S2), and we found that the bioleaching of chalcopyrite by the wild type cells was increased or decreased when removing (media replacement) or adding the sulfur metabolites, respectively (Fig. 5, H and I). Therefore, we conclude that the reduced cell attachment coupled with enhanced iron oxidation enabled the improved bioleaching of the pyrite and chalcopyrite.

It is noteworthy that such high chalcopyrite bioleaching efficiencies are not commonly observed with iron-oxidizing bacteria alone, such as *Leptospirillum* sp (5). This phenomenon may be partially attributed to the stronger adhesion forces exhibited by *Leptospirillum* sp., on minerals, in comparison to other bioleaching organisms including *A. ferrooxidans* (44). Rapid cell adhesion by *Leptospirillum ferrophilum* leads to the swift coverage of chalcopyrite surfaces with sulfur and substantial amounts of jarosite, which result in low copper extraction (45). Furthermore, recent studies indicate that a deficiency of sulfur-oxidizing bacteria during bioleaching can result in low mineral dissolutions (46–48). Indeed, a previous bioleaching experiment using dialysis membranes reported that the bioleaching of copper waste materials reached the lowest efficiency when solely relying on the contact mechanism by *A. ferrooxidans* (49). Considering that the dPetB2 cells can still oxidize sulfur, these cells likely overcome the

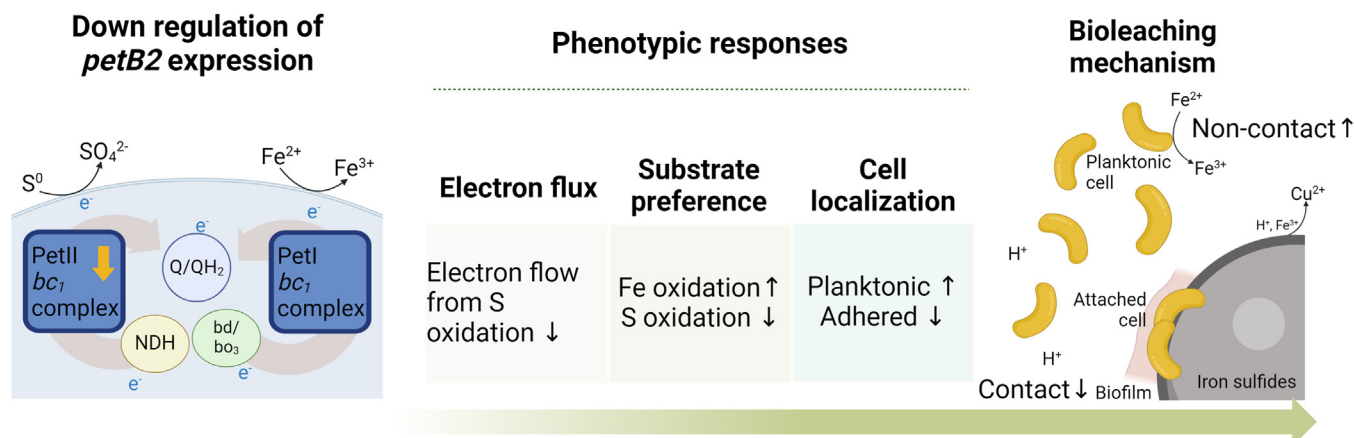


Figure 6. Proposed impacts of *petB2* knockdown on phenotypic shifts of *A. ferrooxidans* and mechanistic consequences during bioleaching. The observed phenotypic responses during the growth and bioleaching, in terms of substrate utilization, cell localizations, and dominant bioleaching mechanisms, which were triggered by the down-regulation of *petB2* gene expression are depicted. The figure was created by [BioRender.com](#).

restricted chalcopryrite bioleaching through accelerated iron oxidation, while maintaining sulfur oxidation to some extent. This advantage could be reinforced by the reduced formation of jarosite. Our observations suggest that an increase in the contribution of the non-contact iron oxidation over the direct contact oxidation mechanism can enhance the bioleaching of iron sulfide ores.

The limited repertoire of genetic tools available for *A. ferrooxidans* has significantly impeded the comprehensive characterization of its unique physiological traits and constrained the exploration of its promising practical applications. Although the CRISPR/Cas system has been widely used for genome editing and silencing, it has been poorly explored in extremophiles due to the harsh growth conditions. Particularly for acidophiles, only two recent studies explored the CRISPRi with dCas9 in *A. ferrooxidans* (22) and *A. ferridurans* (23). We demonstrate the applicability of the CRISPRi/dCas12a in an environmentally and industrially important biomining organism, *A. ferrooxidans*, and explore the phenotypic responses of the core genes involved in the key metabolic processes of this organism. The technical advantages of Cas12a over Cas9 should enable the multiplexing of genome editing, which will accelerate our ability to explore the poorly characterized metabolic pathways of this organism. Furthermore, the compatibility of the based Cas12a/CRISPRi system provides a versatile molecular tool for genetic manipulation of other extremophiles, where genetic modification is often challenging.

Experimental procedures

Experimental design

The objective of this study was to use CRISPRi to attenuate the expression of the electron transport chain *bc₁* complex genes in *A. ferrooxidans* to regulate the flux of electrons from iron or sulfur oxidation towards NADH regeneration or oxygen reduction. Impacts on substrate utilization and growth were characterized, and the impact of the *petB2* knockdown on mineral bioleaching was investigated. This provided new insights the bioleaching mechanisms, including the importance of non-contact interactions and the inhibitory effects of sulfur intermediates.

Experimental materials

Strains used in this study include *E. coli* DH10 β and *E. coli* BL21 obtained from NEB (Ipswich, MA), *E. coli* S17-1 ATCC 47055, and *A. ferrooxidans* ATCC 23270 purchased from ATCC. All *A. ferrooxidans* strains were initially grown in 100 ml of iron and sulfur growth media (F2S medium) with an initial optical density measured at 600 nm (OD₆₀₀) of 0.001, which corresponds to a cell density of 8.3×10^6 cells/ml (50), in shaking incubator (30 °C and 140 rpm). The F2S medium consisted of (NH₄)₂SO₄, 0.8 g/L; HK₂PO₄, 0.1 g/L; MgSO₄·7H₂O, 2.0 g/L; Trace mineral solution (MD-TMS, ATCC), 5 ml/L; citric acid, 1.92 g/L; FeSO₄·7H₂O, 27.8 g/L; and dispersed sulfur (#S789400, Toronto Research Chemicals), 0.1% (w/v). The media was filtered through a 0.2 μ m pore size (Thermo Fisher Scientific) prior to use, and sulfur was added to the media after filtration.

The pFnCpf1_min (pY002) (29) plasmid expressing FnCpf1 (Cas12a) and spacers 1 to 4 of CRISPR array following the J23119 promoter, from *Francisella tularensis* subsp. *Novicida*, was sourced from Addgene (#69975).

Enzymes and reagents for DNA manipulation were obtained from NEB, and oligonucleotides were purchased from Integrated DNA Technologies. All strains, plasmids, and oligonucleotides used in this study were present in Tables S2–S4.

Pyrite (FeS₂, Cat# 77817) was purchased from Sigma-Aldrich and chalcopryrite (CuFeS₂) concentrate (24.8% Cu, 27.2% Fe, and 30.7% S) was provided by Freeport-McMoRan and described elsewhere (6). All chemicals were sourced from Sigma-Aldrich, unless otherwise noted.

Plasmid construction and genetic manipulation

Given the four native gRNA sequences that are from the original *F. novicida* strain following the J23119 promoter, the sequences up to the end of the first native gRNA (with direct repeat sequence of 5' GTCTAAGAACTTTAAATAATTTCT ACTGTTGTAGAT) were amplified and cloned into pYI11 vector, the empty pJRD vector with *tac* promoter, using pYI28 primers (Table S3) via NEBuilder HiFi DNA Assembly following the manufacturer's instructions. The resulting construct (pJRD_Cas12a) was then converted into a catalytically inactivated form by introducing the E1006A and D917A double mutations in pJRD_Cas12a via Q5 Site-Directed Mutagenesis Kit (NEB), using pYI30 and pYI49 primers, respectively (Table S3). The final plasmid (pJRD_dCas12a) was referred to as dCas12a.

To make the pPetA2, pPetB2, pPetA1, and pPetB1 plasmids for knockdown the *petA2*, *petB2*, *petA1*, and *petB1* genes, respectively, the two target spacer sequences for each target gene (Table S4) with a 19-nt direct repeat sequence (5' AATTTCTACTGTTGTAGAT) (51), which contains the cleavage site to separate the first crRNA from the second crRNA, were inserted into the pJRD_dCas12a by Q5 Site-Directed Mutagenesis Kit (NEB), using relevant primer sets (Table S3). All constructed plasmids were transformed into *E. coli* DH10 β for sequence verification before use. The gRNA sequences were designed using Benchling and CHOPCHOP((52)).

After verification, placY1–Y3 were transformed into *E. coli* BL21 and referred to as lacY1, lacY2, and lacY3, respectively. In addition, the sequence verified pPetA2, pPetB2, pPetA1, and pPetB1 were conjugally transferred to *A. ferrooxidans* by filter mating technique using the donor strain *E. coli* S17-1, as previously described (53). The resulting transconjugants were recovered in AFM1 medium containing 0.8 g/L, (NH₄)₂SO₄; 0.1 g/L, HK₂PO₄; 2.0 g/L, MgSO₄·7H₂O; 5 ml/L, MD-TMS; and 72 mM, FeSO₄·7H₂O (final pH of 1.8) and screened in SM4 selection medium containing 0.8 g/L, (NH₄)₂SO₄; 2.0 g/L, MgSO₄·7H₂O; 0.1 g/L, K₂HPO₄; 0.19 g/L, citric acid; 5 ml/L, MD-TMS; 40 μ g/ml, leucine; 19 μ g/ml, diaminopimelic acid; 17.9 μ g/ml, Fe₂(SO₄)₃; 1 g/L, dispersed sulfur; (final pH of 5.0) with kanamycin (50 mg/ml) to isolate the engineered cells. The isolated strains recombinantly expressing pPetA2, pPetB2,

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pPetA1, and pPetB1 were referred to as dPetA2, dPetB2, dPetA1, and dPetB1, respectively.

Quantification of transcriptional gene expression

The wild type and engineered strains (dPetA2, dPetB2, dPetA1, and dPetB1) were harvested at the stationary phase during the growth under 50 ml of F2S medium. Total RNA of the cells was extracted using RNeasy Mini kit (Qiagen), and reversely transcribed to cDNA using QuantiTect Reverse Transcription Kit (Qiagen), as per the protocols. The concentration of *petA2*, *petB2*, *petA1*, and *petB1* genes were analyzed by QIAcuity Digital PCR (dPCR) System (Qiagen). In addition, the expressions of genes responsible for biofilm formation were measured after the bioleaching experiments. 1 ml of cell-mineral mixtures was collected from each condition of the bioleaching. The total RNA was extracted, then converted to cDNA, as described previously (54). Several genes that were proposed to encode the extracellular polymeric substance (EPS) precursors UDP-glucose, UDP-galactose, and dTDP-rhamnose (*luxA*, AFE_2321; *pgm*, AFE_2324; *galU*; and AFE_2323) (55), and genes encoding two distinct signaling mechanisms of quorum sensing (*afeI*, AFE_1999) (54) and c-di-GMP (AFE_0053, AFE_1172, AFE_1360, AFE_1373, AFE_1374, AFE_1379, and AFE_1852) (56), which regulate the biofilm formation, in *A. ferrooxidans*, were selected. The template cDNA was fragmented by *XbaI* and the dPCR reactions were carried out with the QIAcuity EG PCR Kit (Qiagen), following the manufacturer's instructions. The absolute copy number of genes in samples was calculated using Poisson statistics and the final transcriptional gene expression was normalized with cDNA concentration of sample. The primer sequences for the dPCR reactions were present in Table S3.

Phenotypic evaluation of CRISPRi

The effect of the different CRISPR arrays (*i.e.*, number of gRNA and secondary structure) on knockdown efficiency was evaluated in *E. coli* cells. The *lacY1*, *lacY2*, and *lacY3* cells were grown in M9 medium (57) with 0.4% lactose to limit the cell to use lactose for growth, while the lactose induces the *tac* promoter to express our dCas12a systems. The growth of cells with different CRISPR array were monitored, in terms of OD measured at 600 nm.

The phenotypic responses of the engineered *A. ferrooxidans* by silencing *petA2* and *petB2* genes were examined by the growth tests with different sulfur concentrations. The dPetA2 and dPetB2 cells were grown in 100 ml of F2S medium with low (0.1%, w/v) and high (0.5%, w/v) dispersed sulfur. The profiles of iron and sulfur oxidations of the conditions were monitored, in comparison with the wild type, dPetA1, and dPetB1 cells. For dPetA2 and dPetB2 cells, the growth in the absence of sulfur (F2S medium without sulfur) was additionally assessed. The growth tests with *A. ferrooxidans* were performed in a shaking incubator (30 °C and 140 rpm), in triplicate biological replicates.

Bioleaching experiments

Bioleaching experiments were conducted to see if the improved iron oxidation by repressing the gene involved in the

sulfur oxidative electron transfer pathway can affect the bioleaching of the sulfidic ores. Since we observed that the downregulating effects for the tested genes were more obvious in dPetB2 cells than dPetA2, the bioleaching experiments were performed only with the dPetB2, in comparison with the wild-type and dPetB1 cells.

Bioleaching experiments of pyrite and chalcopyrite were conducted in 250 ml Erlenmeyer flasks with working volumes of 100 ml. The leaching media consisted of 0.8 g/L, (NH₄)₂SO₄; 0.1 g/L, K₂HPO₄; 2.0 g/L, MgSO₄·7H₂O; 5 ml/L, MD-TMS, and 10 mM, citric acid (final pH of 1.8). For chalcopyrite bioleaching, 100 mM of FeSO₄·7H₂O was added to accelerate initial leaching due to the poor bioleaching efficiency of this mineral (6, 42, 43). Wild type or engineered cells (dPetB2 or dPetB1) at OD₆₀₀ of 1.0 were inoculated to adjust the initial cell density of 8.3×10^7 cells/ml. In addition, the effect of the presence of sulfur intermediates on the bioleaching was further examined either by removing or adding sulfur intermediates during the bioleaching of chalcopyrite with the wild type cells. The wild type cells at an initial OD₆₀₀ of 0.01 were inoculated to a 14 ml Falcon round-bottom tube with a 10 ml working volume (the same leaching media used for the chalcopyrite bioleaching above). For removing the sulfur metabolites, the leaching solution was periodically replaced (once in 1–3 days) with fresh media. For adding sulfur intermediates, thiosulfate (sodium thiosulfate) tetrathionate (sodium tetrathionate dihydrate), which are the common sulfur metabolites produced during the bioleaching of sulfidic ores, were exogenously added at the final concentrations of 50 µM. The conditions without removing or adding sulfur intermediates were operated as controls. For all bioleaching experiments, the pulp densities of pyrite or chalcopyrite of 1.0% (w/v) were used. All experimental conditions were run in triplicate biological replicates and incubated at 30 °C and 140 rpm. The water evaporation was compensated by adding distilled water, and pH was not adjusted as it was maintained below two for all tested conditions throughout the experiments.

Analytical methods

OD₆₀₀ was measured using a GENESYS 10S UV-VIS spectrophotometer. The soluble Fe²⁺ concentrations were measured by titration with cerium sulfate with a ferroin indicator, and sulfate concentration was measured using a barium sulfate turbidimetric method, as previously described (6, 58). After the bioleaching experiments, total soluble iron and copper concentrations were analyzed by atomic absorption spectrometer (iCE 3300, Thermo Fisher Scientific, Waltham, MA). The solid residues after the bioleaching were characterized with a PANalytical XPert3 Powder X-ray diffraction (XRD) with Empyrean Cu Kα radiation ($k = 0.15418$ nm) equipped with a PIXcel1D detector, and a Zeiss-Sigma VP scanning electron microscopy (SEM) connected to a Bruker XFlash Detector. The sample preparations for XRD and SEM were described previously (6). Biofilm formation was quantified using crystal violet assay, following the previous description (6). For quantifying the number of planktonic cells, 1 ml

of cell-mineral mixtures were filtered through a membrane filter with 0.7 µm pore size, then the filter-through was mixed with 5× SYBR Green I nucleic acid stain (Invitrogen, USA), according to the previous method (50).

Statistical analysis

Statistical significance was determined in Figures 2–5, S1 and Table 1 by one-way ANOVA ($p < 0.05$). All experiments were performed in at least triplicate.

Data availability

All data is contained with the manuscript and Supporting information.

Supporting information—This article contains supporting information (61).

Author contributions—Y. I., S. B., and H. J. writing—review & editing; Y. I. and H. J. methodology; Y. I., S. B., and H. J. formal analysis; Y. I., S. B. and H. J. conceptualization; S. B. supervision; S. B. project administration; S. B. funding acquisition; H. J. writing—original draft; H. J. investigation; H. J. data curation.

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Abbreviations—The abbreviations used are: CRISPRi, CRISPR interference; CRISPR, Clustered Regularly Interspaced Short Palindromic Repeats; RISCs, reduced inorganic sulfur compounds.

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